

Python Course for Physicists (And everyone else!)

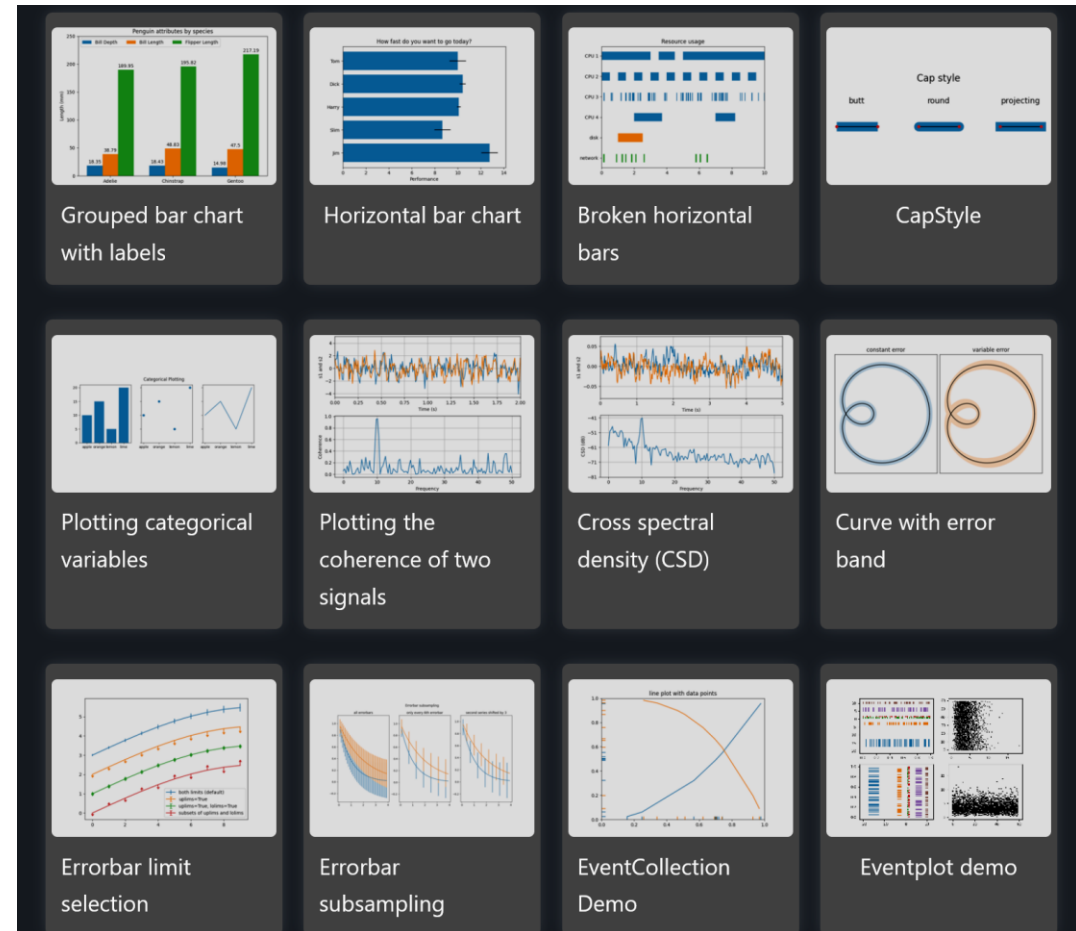
Chunk 9: Matplotlib

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Visualizing your Data: Matplotlib

- Most used plotting and visualization library
- [Examples](#)



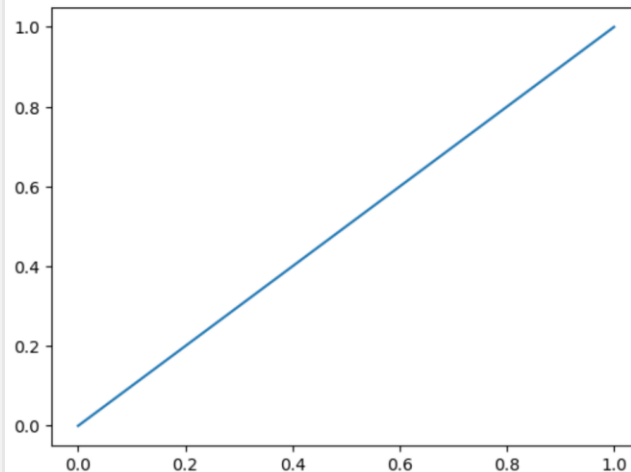
Live Coding: Getting Started

- Anaconda Powershell Prompt: conda install matplotlib (See chunk 7 – imports for details)
- Restart Kernel (if used before)
- `import matplotlib.pyplot as plt`
- Try:

submodule

submodule

```
import matplotlib.pyplot as plt  
plt.plot([0, 1])  
plt.show()
```



Two Types of Plotting

- Two ways to use matplotlib: Quick & dirty or the clean way.
- Quick & dirty:
 - Less code
 - Faster to learn
 - Limited
- Clean way:
 - A little more boilerplate code
 - Slightly different coding “style”
 - Better for complex plotting (e. g. multiple plots next to each other)
- Problem: Codes are similar, but slightly different
→ Be aware when googling stuff

```
import matplotlib.pyplot as plt

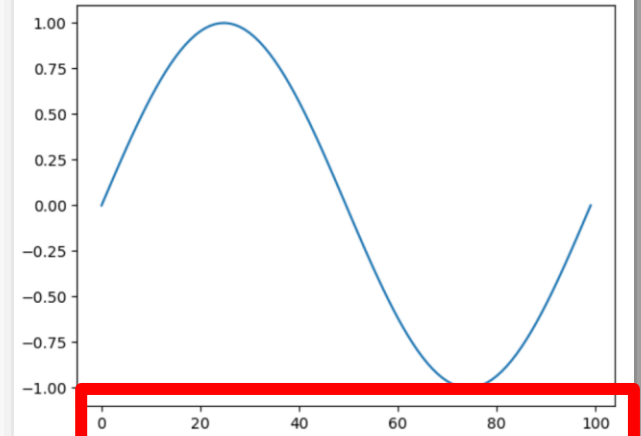
plt.plot([0, 1])
plt.xlabel("x")

fig, ax = plt.subplots()
ax.plot([0, 1])
ax.set_xlabel("x")
```

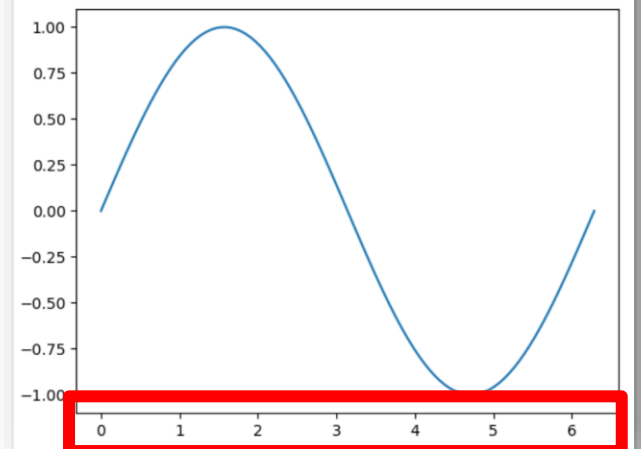
Live Coding: Basics of Plotting

- `plt.plot()`: Can be used in two ways
 - Passing y-data → x-Axis is the index
 - Passing x- and y-data: → normal axis
- `plt.show()`: Generates the plot
 - Mostly works without `plt.show()` but good practice to use it.

```
x = np.linspace(0, 2 * np.pi, 100)  
y = np.sin(x)  
plt.plot(y)  
plt.show()
```



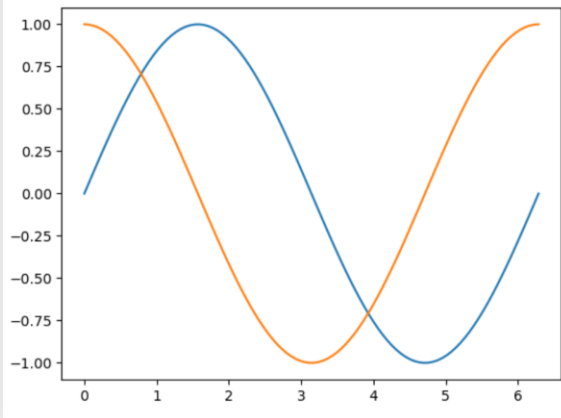
```
x = np.linspace(0, 2 * np.pi, 100)  
y = np.sin(x)  
plt.plot(x, y)  
plt.show()
```



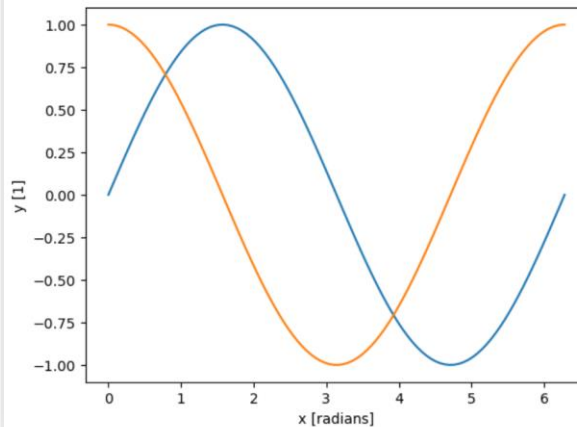
Live Coding: Adding to your Plot

- More data: repeat `plt.plot()`
- Add x- and y-label to your plot: `plt.xlabel("x")` and `plt.ylabel("y")`
- Add a background grid: `plt.grid()`

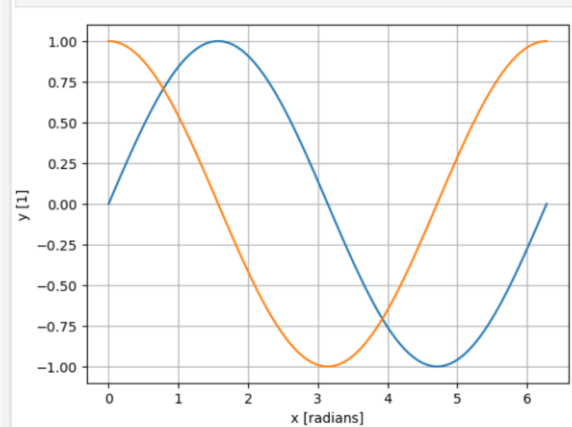
```
x = np.linspace(0, 2 * np.pi, 100)
y1 = np.sin(x)
y2 = np.cos(x)
plt.plot(x, y1)
plt.plot(x, y2)
plt.show()
```



```
x = np.linspace(0, 2 * np.pi, 100)
y1 = np.sin(x)
y2 = np.cos(x)
plt.plot(x, y1)
plt.plot(x, y2)
plt.xlabel("x [radians]")
plt.ylabel("y [1]")
plt.show()
```



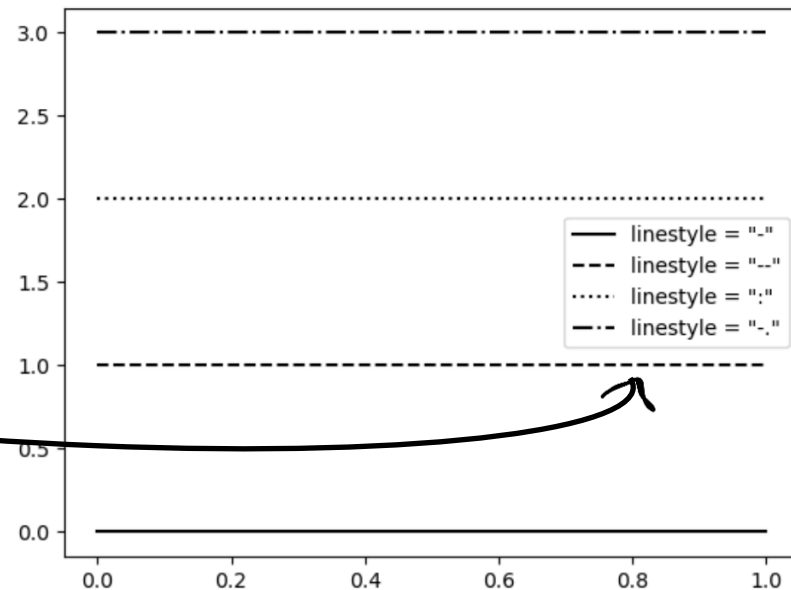
```
x = np.linspace(0, 2 * np.pi, 100)
y1 = np.sin(x)
y2 = np.cos(x)
plt.plot(x, y1)
plt.plot(x, y2)
plt.xlabel("x [radians]")
plt.ylabel("y [1]")
plt.grid()
plt.show()
```



Options for plt.plot()

- Besides x- and y-data, we can pass more optional parameters to the plot() function. To specify what we are passing, we can use “=”.
- Example: plt.plot(x, y, c=“lightgrey”)
- Optional arguments:
 - c → Color of the graph
 - linestyle → Style of the line
 - label → Strings get imaged

→ Need to call legend()



Good Luck with your Tasks 😊

